

Unified Consciousness Theory II — Publication Supplement

Author: Hongju Liu

Affiliation: Independent researcher, Shenzhen, China

Report: TA-TR-2026-21

Version: 1.0

Date: 28 September 2026

DOI: 10.5281/zenodo.23008262

Status: Publication supplement; not peer reviewed.

1. Publication posture

- English theoretical preprint in the Unified Consciousness Theory series.
- Depends on *Unified Consciousness Theory I: From Experience Existence to Experiential Structure* (TA-TR-2026-20, DOI 10.5281/zenodo.23005588).
- Substantial ChatGPT (OpenAI GPT-5.6 Sol) assistance under human direction is disclosed in the main manuscript.
- This is adjacent first-party research. It does not define, amend, validate, or authoritatively interpret the Trinity Accord or its Bitcoin Originals.
- DOI registration, repository integration, OTS, Bitcoin attestation, Arweave preservation, or public readback establish identity and availability of a version; they do not establish truth, originality, significance, peer review, or indexing.

2. Main contribution

Paper II asks whether operational substructures of major consciousness theories can be reconstructed, embedded, reinterpreted, or shown to conflict inside the physically anchored UCT-I mother formalism without collapsing formal recovery into ontological endorsement or empirical confirmation.

The release contains four methodological layers:

1. **Bridge admissibility** — BAC-1 through BAC-7 block output-oracular, per-instance, post-hoc translators.
2. **Translation fidelity** — target-specific signatures, strong claims, explicit residual commitments, and failure conditions are retained.

3. **Two-axis classification** — formal recovery F0-F4 is separated from empirical validation V0-V4; REC/EMB/REI/CON are relation types rather than evidence grades.
4. **Shared-law proof of concept** — two fixed-parameter toy models generate multiple theory-linked organization observables with negative controls and explicit dissociations.

3. Formal-logic audit

The v1.0-rc4 dependency audit encoded 43 logical nodes and 102 directed dependency edges.

- Directed acyclic graph check: **PASS**.
- Directed cycles detected: **0**.
- No atlas result or toy-model result feeds back into A1, A3, A5, or the definition of the UCT-I mother structure.

This audit does not prove that no conceivable objection exists; it records that no fatal internal circularity or contradiction was found in the encoded core argument.

4. Nine-theory atlas

The formal target atlas contains exactly nine theories:

1. IIT 4.0;
2. Recurrent Processing Theory;
3. Global Neuronal Workspace Theory;
4. Higher-Order Theories;
5. Predictive Processing / Active Inference;
6. Dendritic Integration Theory;
7. Memory Theory of Consciousness;
8. Attention Schema Theory;
9. Temporospatial Theory of Consciousness.

ICT, IWMT, TICL and other integrative approaches are comparison/prior-art frameworks, not additional entries in the nine-theory atlas.

5. Claim boundaries

The paper does **not** claim:

- that UCT proves all nine target theories;
- that UCT reduces all nine target theories;
- that formal reconstructibility proves target-theory truth;
- that Paper II empirically proves UCT-I A1 or A5;
- that robust full-theory compression has been established;

- that V3 held-out human/animal recovery has been achieved;
- that V4 replicated cross-system recovery has been achieved;
- that the two toy models constitute biological validation;
- that UCT is the first framework to integrate or compare consciousness theories.

6. Toy-model reproducibility boundary

The original historical model-generation script was not found among the preserved research artifacts. A release-side reconstruction was therefore built from the archived equations and output tables.

The release-side implementation reproduces the six core numerical CSV outputs to less than $1e-9$ maximum absolute discrepancy. It supports the reported structural property that a fixed parameterization is shared across the theory-linked outputs in each toy model.

The prepublication freeze also includes `uct-ii-audit-bundle-v1.0.zip`, containing the formal dependency map, nine-theory relation ledger, source-trace/compression records, model specifications, archived CSV outputs, and release-side reconstruction materials used in the final audits.

It does **not** independently certify the historical chronology of parameter selection or claim byte identity with the unavailable original script.

The toy-model evidence class is therefore:

$$F3/V1$$

rather than V3/V4 empirical recovery.

7. Compression result

The paper demonstrates shared formal reuse across the target-theory translations, but a symbolic description-length stress test changes sign under conservative shared-core and bridge-cost assumptions.

Therefore:

shared formal reuse: supported

while:

robust full-theory compression: not established

8. Strongest remaining scientific burden

The next major scientific step is not to add more theory names. It is to preregister fixed bridge laws and test them on held-out empirical data against relevant baselines, with failure treated as a real possible outcome.

Dependency record

UCT II depends on UCT I for the basal UCT ontology and mother formalism. UCT II does not amend the published UCT-I v1.0 record.